



Overview of The Inaugural Space Health Impacts for the NASA Experience (SHINE) Training Program - Virtual Space Radiation Curriculum



Janice A. Zawaski¹, S. Robin Elgart², Sigrid Reinsch³

¹ National Aeronautical and Space Administration (NASA) Johnson Space Center, ² University of Houston, and
³ National Aeronautical and Space Administration (NASA) Ames Research Center

ABOUT

The Space Radiation Element part of NASA Human Research Program initiated a virtual, annual space radiation curriculum. The SHINE space radiation curriculum aims to educate participants not only in the scientific aspects of space radiation but also in the agency's risk management strategies. SHINE combines weekly seminars by speakers from NASA, other government agencies, academia, and industry, with networking sessions designed to foster collaboration between participants and interactions with NASA scientists. A total of 59 applications were received from which 25 participants were selected. The 25 participants, with citizenship from a total of 8 countries, are composed of 5 graduate students, 4 postdocs, 11 scientists and 5 professors/medical doctors. Participants have a wide range of expertise including molecular and cellular biology, microbiology, botany, physiology, engineering, physics, aerospace medicine, planetary science, biostatistics, and modeling. The inaugural course ran weekly from February 2023 to August 2023 and was comprised of lectures, less formal coffee hours and office hours. The topics covered ranged from space environments to health effects to granting opportunities. The second annual SHINE training program is scheduled from February to August 2024. In addition, a separate competitive practicum session, to allow participants to gain hands-on radiation experience, will be held Fall 2024 at the NASA Space Radiation Laboratory (NSRL).

Prior Syllabus (2023)

Month	Type	Topic	Speakers/Affiliation
February	Seminar	Space Radiation Element	Robin Elgart, NASA JSC Greg Nelson, KBR, Loma Linda University Janapriya (JP) Saha, Leidos, Inc. Brock Sishc, Aegis Aerospace
	Seminar	Radiation Physics and Dosimetry	Cary Zeitlin, Leidos, Inc.
March	Coffee Hour	HRP/element (breakout room)	Jancy McPhee, NASA JSC
	Seminar	Space Environment	Kerry Lee, NASA JSC
	Coffee Hour	Protons/Therapy	Marco Durante, GSI Helmholtzzentrum für Schwerionenforschung
	Seminar	Space Radiation Analogs (NSRL and others)	Tony Slaba, NASA Peter Guida, Brookhaven National Laboratory
April	Coffee Hour	STAR/Space Biology	Lisa Carnell, NASA HQ Egle Cekanaviciute, NASA ARC
	Seminar	Mechanism of Damage and Repair	Anthony (Tony) Davis, University of Texas Southwestern Claudia Wiese, Colorado State University
	Office Hours	Mock Beam Time proposal	HRP element only
	Seminar	Mechanism of Damage and Repair – Oxidative Stress	Kathryn (Kathy) Held, National Council on Radiation Protection and Measurement/ Massachusetts General Hospital/Harvard Medical School
May	Coffee Hour	HSRB overview	Mary Van Baalen, NASA JSC Jerry Shay, University of Texas Southwestern
	Seminar	Biomarkers	Catherine (Katie) Davis-Takacs, Uniformed Services University of the Health Sciences Candice Tahimic, University of North Florida
	Office Hours	Mock Beam Time proposal	HRP element only
	Seminar	Radioprotectants & Mitigation	Michael Story, University of Texas Southwestern Marjan Boerma, University of Arkansas for Medical Sciences
June	Coffee Hour	NASA Open Science - Genelab, LSDA etc.	Gregory Nelson, Loma Linda University, KBR America Reyes Wang, KBR Inc. Sigrid Reinsch, NASA ARC Jessica Keune or Mary Van Baalen, NASA JSC Mike Weil (cancer), Colorado State University, Chris Porada (various Tissue Chips), Wake Forest University
	Seminar	Experimental Models I	David Goukassian , Icahn School of Medicine Joe Wu , Stanford
	Seminar	Experimental Models II	CNS models Egle Cekanaviciute , NASA Amelia Eisch/Fred Kiffer, University of Pennsylvania Cynthia (Cindy) Lemere, Brigham and Women's Hospital, Harvard Medical School
	Office Hours	HRP element only	HRP element only
	Seminar	Radiation epidemiology & Biostatistics	Cato Milder, NIH Lori Chappell, KBR Inc.
July	Seminar	Bioinformatics & Data Modeling	Parastou Eslami, Universal Artificial Intelligence Prachi Kothiyal, Scimentis
	Coffee Hour	Grants and Funding Opportunities	Ben Goodman, NASA Research and Educational Support Services (NRESS) Janice Zawaski, NASA Sharmila Bhattacharya, NASA
	Presentations	Mock Beamline Proposals	Participants
August	Seminar	Coffer Hour and Closing Session	Mary Van Baalen, NASA

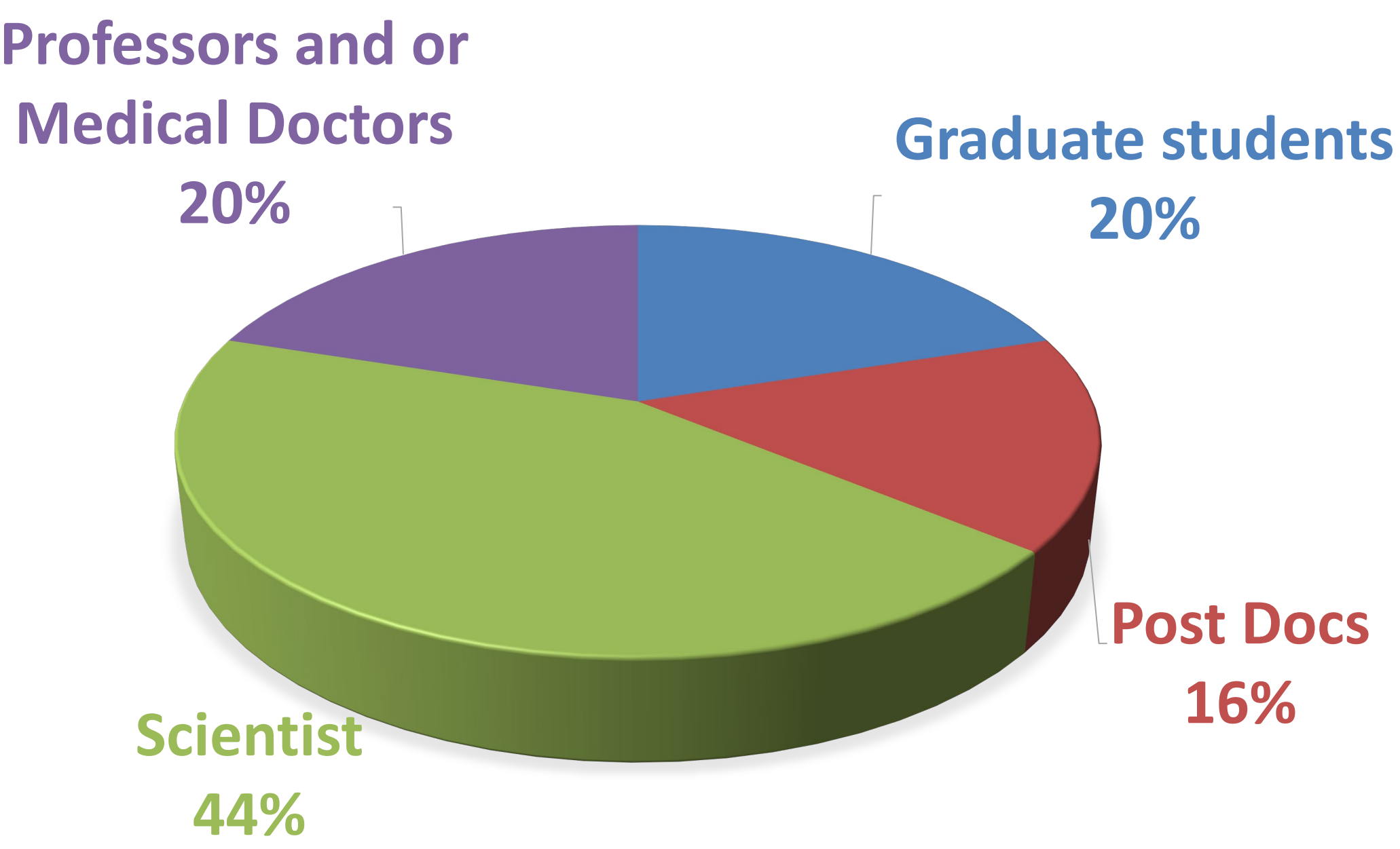
SPEAKERS



PARTICIPANTS

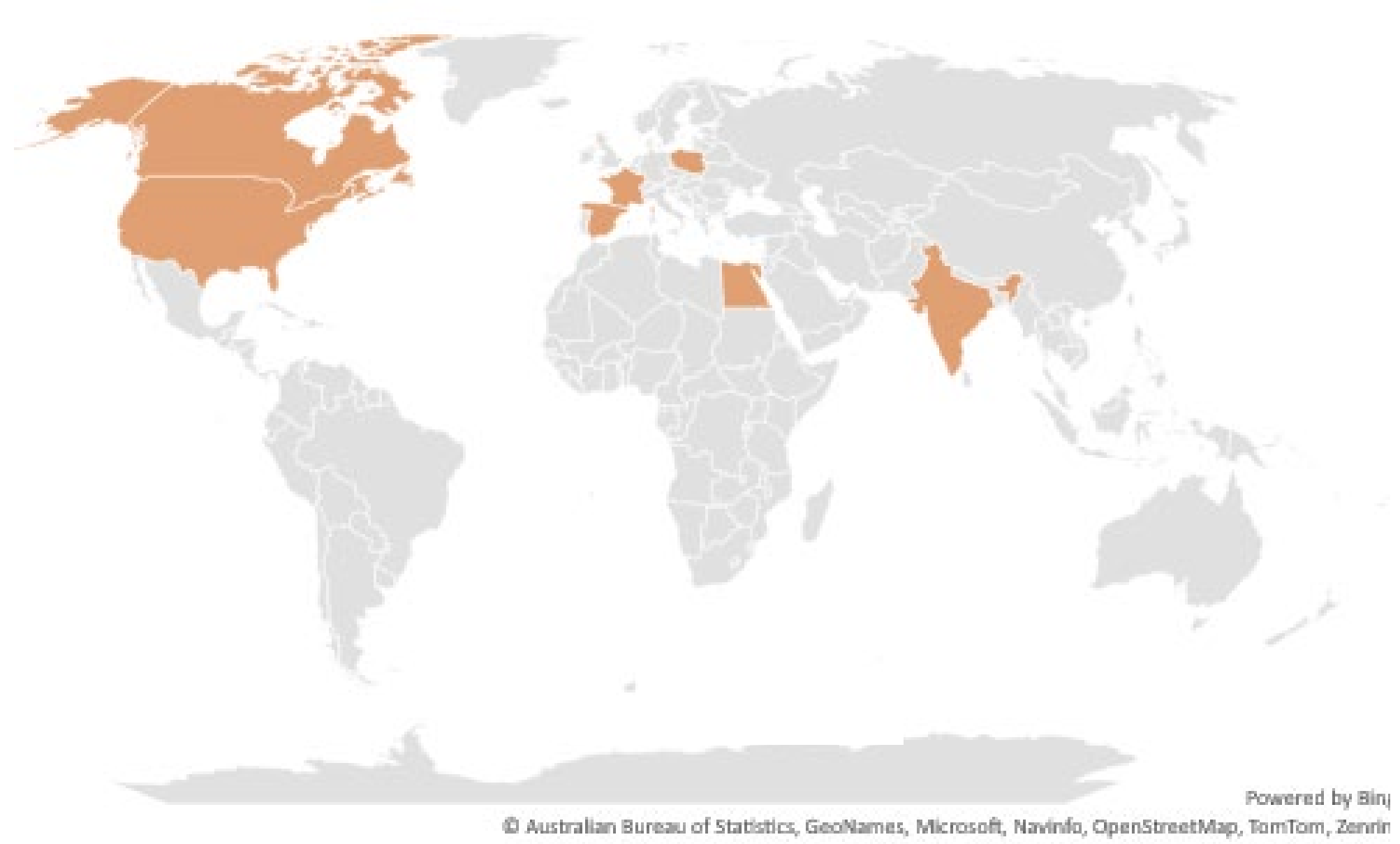


SHINE 2023 COMPOSITION

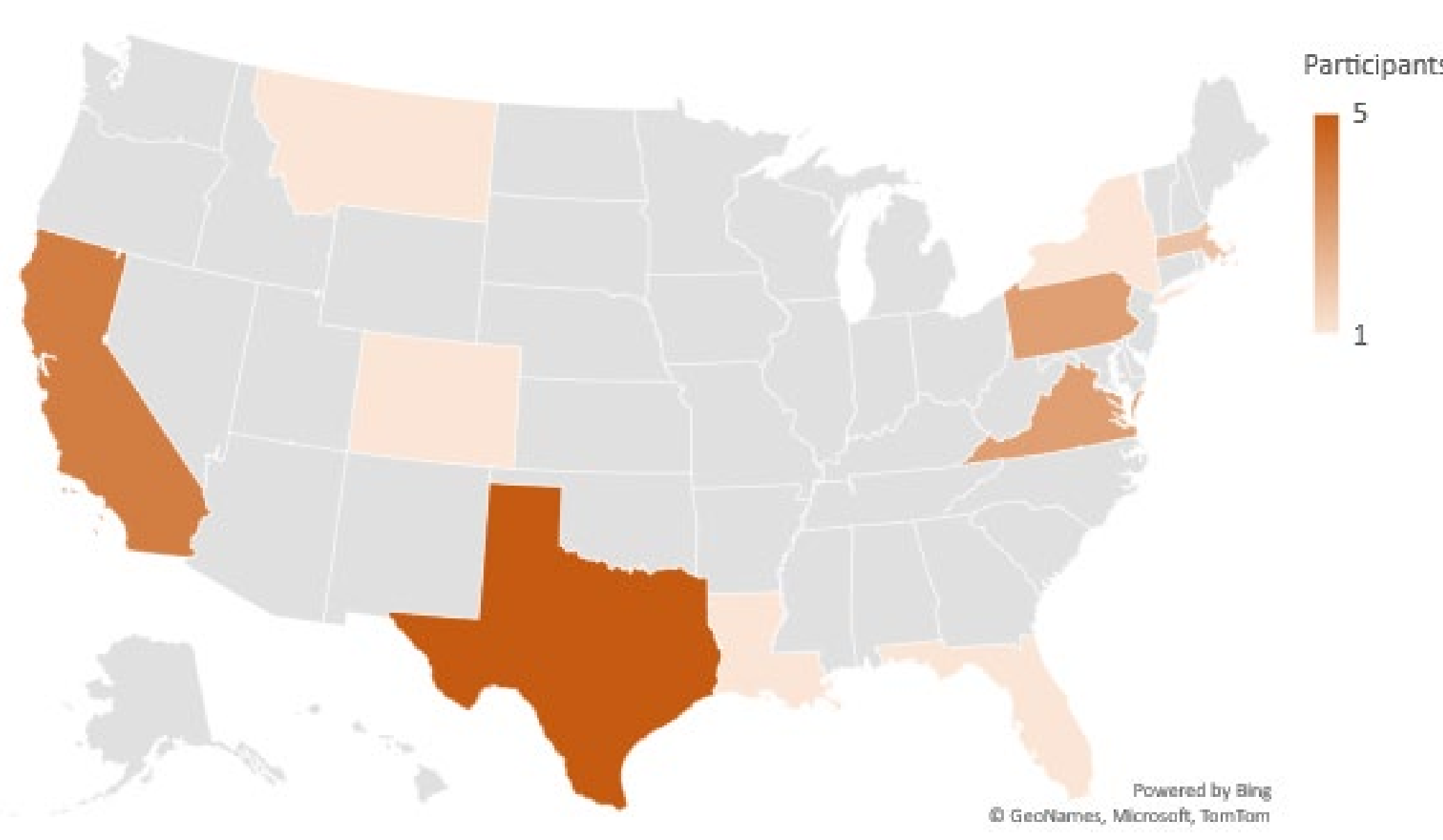


Participant Distribution

Citizenship Participation



Geographical Distribution of Participants



More Information

Contact us for more information jsc-hrp-space-radiation-element@mail.nasa.gov



2024 SR
Didactic
SHINE

2024 SR
Practicum
SHINE